

AI Assisted Coding (III Year) Assignment

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Lab 16 – Database Design and Queries (AI Assisted)

Lab Objectives

- **Use AI to assist in designing and implementing fundamental data structures in Python**
- **Learn prompt-based structure creation, optimization, and documentation**
- **Improve understanding of Lists, Stacks, Queues, Linked Lists, Graphs, and Hash Tables**
- **Enhance readability and performance with AI-generated suggestions**

Task 1 – Student Information System Schema

1. AI Prompt Used

Generate SQL schema for a Student Information System.

Tables required:

Students

Courses

Enrollments

Requirements:

- Define primary keys
- Define foreign key relationships
- Maintain referential integrity
- Generate SQL queries for:
 - Insert a new student
 - Fetch all courses enrolled by a student
 - Count students in each course

CODE and OUTPUT:

```

MySQL
INSERT INTO Course (course_id, course_name)
VALUES (1, 'Database Systems');

* sqlite:///lab08.db
1 row affected.
[]

MySQL
INSERT INTO Enrollments (student_id, course_id)
VALUES (1, 101);

* sqlite:///lab08.db
1 row affected.
[]

MySQL
CREATE TABLE Students (
  student_id INTEGER PRIMARY KEY,
  name TEXT NOT NULL,
  email TEXT UNIQUE
);

CREATE TABLE Courses (
  course_id INTEGER PRIMARY KEY,
  course_name TEXT NOT NULL
);

CREATE TABLE Enrollments (
  enrollment_id INTEGER PRIMARY KEY,
  student_id INTEGER,
  course_id INTEGER,
  FOREIGN KEY(student_id) REFERENCES Students(student_id),
  FOREIGN KEY(course_id) REFERENCES Courses(course_id)
);

-- * sqlite:///lab08.db
Done.
Done.
Done.
[]

MySQL
INSERT INTO Students (student_id, name, email)
VALUES (1, 'Akshay', 'akshay@email.com');

* sqlite:///lab08.db
1 row affected.
[]

import pandas as pd
df = sql --dataframe SELECT Students.name, Courses.course_name FROM Enrollments JOIN Students ON Students.student_id = Enrollments.student_id JOIN Courses ON Courses.course_id = Enrollments.course_id WHERE Students.student_id = 1;
display(df)

* sqlite:///lab08.db
None

import pandas as pd
df = sql --dataframe SELECT Courses.course_name, COUNT(Enrollments.student_id) AS total_students FROM Enrollments JOIN Courses ON Courses.course_id = Enrollments.course_id GROUP BY Courses.course_name;
display(df)

-- * sqlite:///lab08.db
None

```

Students

student_id	name	email
1	Akshay	akshay@email.com ↗
2	Ravi	ravi@email.com ↗

Courses

course_id	course_name
101	Database Systems
102	Machine Learning

Enrollments

enrollment_id	student_id	course_id
1	1	101
2	1	102
3	2	101

Task 2 – Hospital Management Database
AI Prompt

Generate SQL schema for a Hospital Management System.

Tables:

Doctors

Patients

Appointments

Requirements:

- Primary keys
- Foreign keys
- Date validation
- Queries for:
 - List appointments for a doctor
 - Retrieve patient history
 - Count patients treated by each doctor

Schema AND Output:

```
0 CREATE TABLE Patients (  
  patient_id INT(10) PRIMARY KEY,  
  name TEXT,  
  age INT(5)  
);  
  
CREATE TABLE Appointments (  
  appointment_id INT(10) PRIMARY KEY,  
  doctor_id INT(10),  
  patient_id INT(10),  
  appointment_date DATE,  
  FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id),  
  FOREIGN KEY (patient_id) REFERENCES Patients(patient_id)  
);  
  
-- * sqlite:///lab08.db  
Doc:  
Name:  
Data:  
[]  
  
import pandas as pd  
df = pd.read_csv('SELECT Doctors.name, Patients.name, appointment_date FROM Appointments JOIN Doctors ON Doctors.doctor_id = Appointments.doctor_id JOIN Patients ON Patients.patient_id = Appointments.patient_id WHERE Doctors.doctor_id = 1;  
display(df)  
  
* sqlite:///lab08.db  
Name:  
  
What:  
Insert INTO Doctors (doctor_id, name, specialization) VALUES (1, 'Dr. Smith', 'Cardiology');  
  
* sqlite:///lab08.db  
Columns:long(integer) WHERE constraint failed: Doctors.doctor_id  
[SQL] INSERT INTO Doctors (doctor_id, name, specialization) VALUES (1, 'Dr. Smith', 'Cardiology');  
(Background on this error #: 'sqlite/311616000/20190511')  
  
import pandas as pd  
df = pd.read_csv('SELECT Patients.name, appointment_date, Doctors.name FROM Appointments JOIN Patients ON Patients.patient_id = Appointments.patient_id JOIN Doctors ON Doctors.doctor_id = Appointments.doctor_id WHERE Patients.patient_id = 1;  
display(df)  
  
* sqlite:///lab08.db  
Name:  
  
import pandas as pd  
df = pd.read_csv('SELECT Doctors.name, COUNT(Patients.patient_id) AS total_patients FROM Appointments JOIN Doctors ON Doctors.doctor_id = Appointments.doctor_id JOIN Patients ON Patients.patient_id = Appointments.patient_id GROUP BY Doctors.name;  
display(df)  
  
-- * sqlite:///lab08.db  
Name:  
  
import pandas as pd  
dfDoctors = pd.read_csv('SELECT * FROM Doctors;  
display(dfDoctors)  
  
* sqlite:///lab08.db  
Name:
```

Sample Data

Doctors

doctor_id	name	specialization
1	Dr. Sharma	Cardiology
2	Dr. Rao	Neurology

Patients

patient_id	name	age
1	Arjun	30
2	Meera	25

Appointments

appointment_id	doctor_id	patient_id	appointment_date
1	1	1	2026-03-10
2	1	2	2026-03-11

Task 3 – Library Database

AI Prompt

Generate SQL schema for a Library Management System.

Tables:

Books

Members

Loans

Requirements:

- Add indexing strategy
- Generate queries:
 - Books currently issued
 - Overdue books (loan > 30 days)
 - Books loaned per member

CODE and OUTPUT:

```
CREATE TABLE Books (
  book_id INT PRIMARY KEY,
  title VARCHAR(100) NOT NULL,
  author VARCHAR(100) NOT NULL
);

CREATE TABLE Members (
  member_id INT PRIMARY KEY,
  name VARCHAR(100) NOT NULL
);

CREATE TABLE Loans (
  loan_id INT PRIMARY KEY,
  book_id INT,
  member_id INT,
  loan_date DATE,
  FOREIGN KEY (book_id) REFERENCES Books(book_id),
  FOREIGN KEY (member_id) REFERENCES Members(member_id)
);

INSERT INTO Books (book_id, title, author) VALUES
(1, 'AI Basics', 'John'),
(2, 'SQL Mastery', 'David');

INSERT INTO Members (member_id, name) VALUES
(1, 'Akshay'),
(2, 'Ravi');

INSERT INTO Loans (loan_id, book_id, member_id, loan_date) VALUES
(1, 1, 1, '2026-02-01');
```

Books			
book_id	title	author	
1	AI Basics	John	
2	SQL Mastery	David	
Members			
member_id	name		
1	Akshay		
2	Ravi		
Loans			
loan_id	book_id	member_id	loan_date
1	1	1	2026-02-01

Task 4 – Online Shopping Database

AI Prompt

Generate SQL schema for an E-commerce platform.

Tables:

Users

Products

Orders

OrderDetails

Queries:

Retrieve all orders by a user

Find most purchased product

Calculate monthly revenue

Also suggest normalization improvements.



CODE and OUTPUT:

```
SQL
CREATE TABLE Users (
  user_id INT(10) PRIMARY KEY,
  name TEXT,
  email TEXT UNIQUE
);

CREATE TABLE Products (
  product_id INT(10) PRIMARY KEY,
  name TEXT,
  price DEC
);

CREATE TABLE Orders (
  order_id INT(10) PRIMARY KEY,
  user_id INT(10),
  order_date DATE,
  FOREIGN KEY(user_id) REFERENCES Users(user_id)
);

CREATE TABLE OrderDetails (
  order_detail_id INT(10) PRIMARY KEY,
  order_id INT(10),
  product_id INT(10),
  quantity INT(10),
  FOREIGN KEY(order_id) REFERENCES Orders(order_id),
  FOREIGN KEY(product_id) REFERENCES Products(product_id)
);

-- sqlsh://localhost
Name:
Date:
Date:
[]

Import schema as pd
df = pd.read_sql('SELECT Users.name, Orders.order_id, order_date FROM Orders JOIN Users ON (Users.user_id = Orders.user_id WHERE Users.user_id = 1);',
  sqlsh://localhost)
Name:
Date:
Date:
Date:
[]

Import schema as pd
df = pd.read_sql('SELECT Products.name, SUM(quantity) AS total_sold FROM OrderDetails JOIN Products ON Products.product_id = OrderDetails.product_id GROUP BY Products.name ORDER BY total_sold DESC LIMIT 1;',
  sqlsh://localhost)
Name:
Date:
Date:
Date:
[]

Import schema as pd
df = pd.read_sql('SELECT strftime("%Y-%m", order_date) AS month, SUM(Products.price * OrderDetails.quantity) AS revenue FROM Orders JOIN OrderDetails ON Orders.order_id = OrderDetails.order_id JOIN Products ON Products.product_id = OrderDetails.product_id GROUP BY month;',
  sqlsh://localhost)
Name:
```

Users			
user_id	name		
1	Akshay		
Products			
product_id	name	price	
1	Laptop	60000	
2	Mouse	500	
Orders			
order_id	user_id	order_date	
1	1	2026-03-01	
OrderDetails			
order_detail_id	order_id	product_id	quantity
1	1	1	1
2	1	2	2

Task 5 – University Examination Database

AI Prompt:

Generate SQL schema for a University Examination System.

Tables:

Students

Subjects

Exams

Results

Queries:

Insert exam results

Retrieve subject-wise marks

Calculate average marks per subject

CODE and OUTPUT:

```
SQL
-- Create Tables
CREATE TABLE Students (
    student_id INT PRIMARY KEY,
    student_name VARCHAR(100)
);

CREATE TABLE Subjects (
    subject_id INT PRIMARY KEY,
    subject_name VARCHAR(100)
);

CREATE TABLE Exams (
    exam_id INT PRIMARY KEY,
    exam_date DATE,
    FOREIGN KEY (subject_id) REFERENCES Subjects(subject_id)
);

CREATE TABLE Results (
    result_id INT PRIMARY KEY,
    student_id INT,
    subject_id INT,
    exam_id INT,
    marks INT,
    FOREIGN KEY (student_id) REFERENCES Students(student_id),
    FOREIGN KEY (subject_id) REFERENCES Subjects(subject_id),
    FOREIGN KEY (exam_id) REFERENCES Exams(exam_id)
);

-- Insert Data
INSERT INTO Students (student_id, student_name) VALUES (1, 'John Doe'), (2, 'Jane Smith'), (3, 'Mike Johnson');
INSERT INTO Subjects (subject_id, subject_name) VALUES (1, 'Mathematics'), (2, 'Science'), (3, 'History'), (4, 'Literature');
INSERT INTO Exams (exam_id, exam_date, subject_id) VALUES (1, '2023-09-01', 1), (2, '2023-09-01', 2), (3, '2023-09-01', 3), (4, '2023-09-01', 4);
INSERT INTO Results (result_id, student_id, subject_id, exam_id, marks) VALUES (1, 1, 1, 1, 85), (2, 1, 2, 1, 78), (3, 1, 3, 1, 92), (4, 1, 4, 1, 88), (5, 2, 1, 2, 75), (6, 2, 2, 2, 82), (7, 2, 3, 2, 80), (8, 2, 4, 2, 70), (9, 3, 1, 3, 88), (10, 3, 2, 3, 79), (11, 3, 3, 3, 85), (12, 3, 4, 3, 72);

-- Query 1: Insert exam results
INSERT INTO Results (result_id, student_id, subject_id, exam_id, marks) VALUES (13, 1, 1, 2, 78), (14, 1, 2, 2, 80), (15, 1, 3, 2, 85), (16, 1, 4, 2, 75), (17, 2, 1, 3, 82), (18, 2, 2, 3, 78), (19, 2, 3, 3, 88), (20, 2, 4, 3, 70), (21, 3, 1, 4, 80), (22, 3, 2, 4, 75), (23, 3, 3, 4, 82), (24, 3, 4, 4, 78);

-- Query 2: Retrieve subject-wise marks
SELECT subject_id, subject_name, exam_id, exam_date, marks
FROM Results
JOIN Subjects ON Results.subject_id = Subjects.subject_id
JOIN Exams ON Results.exam_id = Exams.exam_id
ORDER BY subject_id, exam_id;

-- Query 3: Calculate average marks per subject
SELECT subject_id, subject_name, AVG(marks) AS average_marks
FROM Results
JOIN Subjects ON Results.subject_id = Subjects.subject_id
GROUP BY subject_id, subject_name;
```


Subjects			
subject_id		subject_name	
1		Database	
2		AI	
Results			
result_id	student_id	exam_id	marks
1	1	1	85
2	1	2	90
Query – Subject Wise Marks			
Output			
student	subject	marks	
Akshay	Database	85	
Akshay	AI	90	

Final Analysis

The experiment demonstrated how AI can assist in designing relational database schemas and generating SQL queries. Multiple database systems were modeled including Student Information Systems, Hospital Management Systems, Library Systems, E-commerce platforms, and University Examination Systems. SQL queries demonstrated CRUD operations, joins, and aggregate functions. Normalization principles and indexing strategies were applied to improve data integrity and query performance.